

SAN-D — Company Description

Page 1 — Technical

Who we are

SAN-D is a fabless semiconductor company: we design memory chips; foundries manufacture them. Architecture, cell design, controller, and firmware are ours; wafer fabrication is outsourced — the NVIDIA model.

The product: UM002B universal memory

The UM002B cell is a 2T1C1M structure — two transistors, one capacitor, one hafnium-oxide (HfOx) memristor per cell:

- **Volatile path** — thin-oxide transistor (T_DRAM) + capacitor: the working memory. DRAM-class speed (nanosecond access); stored as charge, so it leaks, needs refresh, and dies with power.
- **Nonvolatile path** — thick-oxide transistor (T_NVM) + HfOx memristor: the shadow. Stores data as resistance (a filament formed or broken in the oxide); needs no power, but is slower to write — so it stays off the daily critical path.

The capacitor does all normal operation; the memristor wakes only on power events.

Operating modes

- **NORMAL** — DRAM-speed access on the capacitor path; memristor idle.
- **SAVE** — on power-fail detection: charge → resistance, copied row by row (NVDIMM-N-like backup).
- **OFF** — zero standby power; state held as resistance.
- **RESTORE** — on power return: resistance → charge copied back; the machine resumes where it left off.

Controller and scope

An on-die controller handles power-fail detection, SAVE/RESTORE sequencing, and ECC; its firmware lives in on-chip ROM/OTP programmed at manufacture. SAN-D's scope: chip, firmware, host interface, BIOS/UEFI enablement, and OS drivers — "it just works in a PC." Deeper OS-level memory/storage unification is ecosystem work over time (the Intel Optane precedent).

Device strategy

The custom HfOx stack is tuned through a JDA (joint development agreement) with the foundry — device targets set on real wafers. The PDK path hardens the design side: real models, rules, and parasitics. The industrial temperature range (−40 to 85 °C) is a design target; all device figures are targets until measured on silicon.

Business model and status

Dual model: sell UM002B chips and license the cell IP. First market: consumer PCs, where one unified pool replaces the DRAM + SSD hierarchy. Pre-silicon work is complete (simulation, device-physics analysis, GDSII layout, manufacturing-readiness review); gated next steps are PDK access, a signed JDA, foundry engagement, and first silicon — stated as plan, not status.

What SAN-D is building, in everyday words

Every computer has two memories: the **desk (RAM)** — fast, where you work, wiped out by a power cut; and the **filing cabinet (storage/SSD)** — permanent but slow, and you must remember to save into it. For fifty years we have shuttled work between the two and pressed Save to survive power cuts.

SAN-D merges them into one desk that never gets cleared — a single pool of memory that is both fast and permanent.

How it survives a power cut

Think of a camera. Beside the fast working part of each memory cell sits a tiny camera (a memristor) that remembers things as electrical resistance — which needs no power to persist. When power fails, the chip photographs the whole desk (SAVE); while power is off, the photographs sit safe using zero electricity; when power returns, it develops them back onto the desk (RESTORE). On ordinary days the cameras sleep — that is why the computer stays fast.

What this means for a normal PC

- **No boot wait** — the PC wakes instantly, exactly as you left it.
- **Never lose unsaved work** — an unsaved document, an un-checkpointed game, fifty browser tabs: a power cut becomes a pause, not a loss.
- **One memory number** — not "8 GB RAM + 512 GB storage," just one pool of universal memory.

Power loss becomes a pause button, not a catastrophe.

The business, plainly

SAN-D designs the chip; specialist factories (foundries) build it — like NVIDIA. It earns two ways: selling its own memory chips, and licensing the technology to others. First stop: everyday PCs.

Where things stand, honestly

The chip is fully designed on computers — circuits simulated, physics analyzed, layout drawn, manufacturing-readiness reviewed — but not yet manufactured. Next: partnerships with a chip factory (their design toolbox, a joint program to perfect the memory material on real wafers), then first real chips. This page describes the destination, not a done deal.